

# When the Earth Moves

Peru shakes! Read about five big earthquakes, giant waves and volcanoes. See real photos. Learn what to do. Then make a safety flyer for your family.

## Inside this book

- 12 chapters, one for each week
- Activities after every chapter
- Peru in Photos
- Two timelines
- Glossary with Spanish
- How am I doing? rubric



Starters edition • Pre-A1

Nordic International School • Grade 3 • Unit 5 • Lima, Peru

Photos inside!

NORDIC INTERNATIONAL SCHOOL • GRADE 3 • UNIT 5

# When the Earth Moves

Earthquakes in Peru

Starters edition • Pre-A1



Fun for Nordic • The Lighthouse Explorers

## Before you start

### BIG QUESTION

**Peru shakes. What do I have to know, and what do I have to have ready, so that it does not catch my family by surprise?**

Peru is a big country. It has a long coast, high mountains and a big forest.

Sometimes the ground in Peru moves. We call it an **earthquake**.

In this book you read about real earthquakes in Peru. You learn what to do. Then you make a **safety flyer** for a family in your street.

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## Meet The Lighthouse Explorers



**Freya**

shows you the new words



**Nico**

makes mistakes — can you help him?



**Astrid**

gives the instructions



**Tomás**

loves action and never stops



**Pip**

is hiding in every chapter.  
Can you find him?

 **Find Pip!** Pip the puffin is hiding at the start of every chapter. Can you find him every time?

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# Sources and Credits

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# How am I doing?

The unit rubric. AD is the highest level.

| I can...                                                                                           | AD                                                                                | A                                                                                             | B                                                              | C                                                                |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------|
| I name natural disasters, safety words, the things in the emergency kit and the parts of the body. | I use them and I explain what one of them means to somebody who does not know it. | I use the words on my own, in sentences, when I am talking about safety.                      | I say most of the words when I see the picture.                | I say a few of the words with help.                              |
| I read a table and a chart about earthquakes in Peru and I say what they show.                     | I notice something in the data that nobody pointed out to me.                     | I ask and answer questions about the data and I compare two earthquakes.                      | I find information when I am asked a direct question.          | I find one piece of information if somebody shows me the column. |
| I tell what happened in a real Peruvian earthquake using the past simple.                          | I tell it in order and I add what happened afterwards.                            | I tell what happened in four sentences, in order and in the past.                             | I write two or three sentences in the past with some mistakes. | I copy sentences in the past.                                    |
| I use must, must not and should to give safety rules and advice.                                   | I explain to somebody else why one is a rule and the other is advice.             | I choose must, must not or should correctly and I give the reason.                            | I write rules but I mix up must and should.                    | I copy a rule that is given to me.                               |
| I organise safety instructions into before, during and after.                                      | I notice when an instruction could go in two moments and I say why.               | I sort them all and I use the three moments to organise my own flyer.                         | I sort most of the actions with a bit of help.                 | I put two of the three moments in the right place.               |
| I write a safety flyer of 90 to 110 words and I present it to other people.                        | My flyer works for somebody who is younger than me, and I prove it.               | I write the three parts, with a heading, an instruction and a drawing each, and I present it. | I write the three parts but they are short or unclear.         | I write some sentences and I show my drawing.                    |

- How am I doing? (Rubric)

- Sources and Credits



# Our Ground Moves



Monday morning at school.



What's this? The floor is moving!

It is Monday morning at school in Lima. The children are reading.

Then the windows shake. The lamp moves. The floor moves too!

“What's that?” asks Nico.

**first-aid kit** (*botiquín*) — a box with plasters, bandages and things to help people who are hurt

**flood** (*inundación*) — a lot of water covering land that is usually dry

**have to** (*tener que*) — it is necessary: we have to leave now

**head** (*cabeza*) — the top part of your body, with your eyes and mouth

**headache** (*dolor de cabeza*) — a pain in your head

**hurt** (*doler / lastimarse*) — to feel pain: my knee hurts

**intensity** (*intensidad*) — how strongly the shaking is felt in one place

**knee** (*rodilla*) — the middle of your leg, where it bends

**landslide** (*deslizamiento / huaico*) — rocks and earth sliding down a hill

**lava** (*lava*) — hot liquid rock that comes out of a volcano

**magnitude** (*magnitud*) — a number that shows how strong an earthquake is

**meeting point** (*punto de reunión*) — the safe place outside where everyone meets after leaving

**must / must not** (*debe / no debe*) — for rules: you must pack water; you must not run

**plate** (*placa tectónica*) — a giant piece of the Earth's outside

**Ring of Fire** (*Cinturón de Fuego*) — the line of coasts around the Pacific with many earthquakes and volcanoes

**safe zone** (*zona segura*) — the safest place to be in an earthquake

**seismograph** (*sismógrafo*) — a machine that measures the shaking of the ground

**shelter** (*refugio / albergue*) — a safe place to stay

**should** (*debería*) — for advice: you should check your backpack

**stomachache** (*dolor de estómago*) — a pain in your stomach

**torch** (*linterna*) — a small light you carry in your hand

**tsunami** (*tsunami / maremoto*) — very big waves caused by an earthquake under the sea

**volcano** (*volcán*) — a mountain where hot rock, gas and ash come out

**whistle** (*silbato*) — a small thing you blow into to make a loud sound

# Glossary

The words of this unit, with Spanish.

**after** (*después*) — at a later time: after the earthquake

**aftershock** (*réplica*) — a smaller earthquake that comes after a big one

**ankle** (*tobillo*) — the part of the body between your leg and your foot

**ash** (*ceniza*) — the grey powder that comes out of a volcano

**avalanche** (*avalancha / aluvión*) — a very large amount of ice, snow and rock falling fast down a mountain

**back** (*espalda*) — the part of your body behind you, from your neck to your bottom

**backache** (*dolor de espalda*) — a pain in your back

**before** (*antes*) — at an earlier time: before the earthquake

**blanket** (*manta*) — a warm cover for your bed or for sleeping outside

**bruise** (*moretón*) — a blue or purple mark on your skin after a knock

**cut** (*corte*) — an opening in your skin that bleeds

**danger** (*peligro*) — something that can hurt you

**drill** (*simulacro*) — practice for what to do in an emergency

**Drop, cover, hold on** (*Agáchate, cúbrete, sujétate*) — what you do during the shaking

**during** (*durante*) — while something is happening

**earthquake** (*sismo / temblor / terremoto*) — a sudden shaking of the ground, big or small: in English it is always “earthquake”

**emergency kit** (*mochila para emergencias*) — a bag with the things you need in an emergency

**epicentre** (*epicentro*) — the place on the ground right above where an earthquake starts

**exit** (*salida*) — the way out of a building

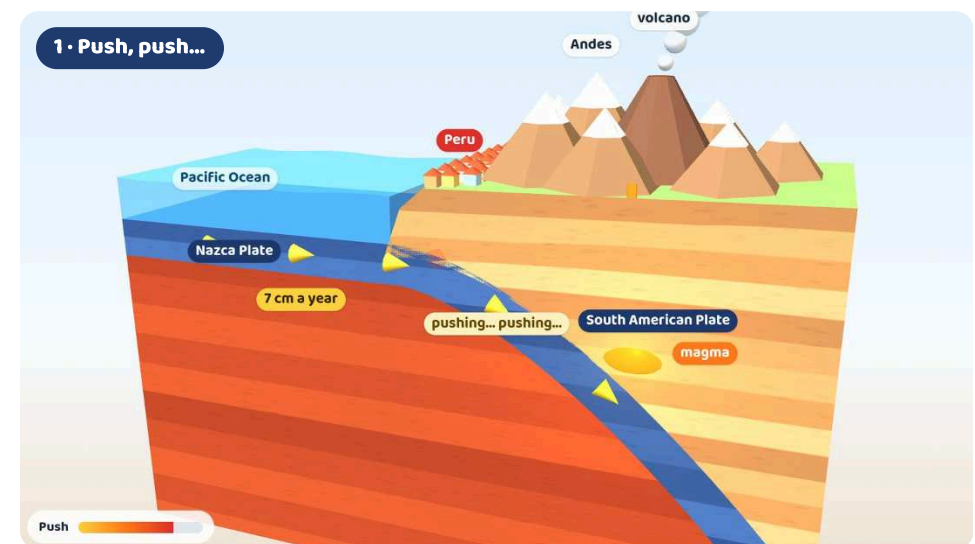
**fault** (*falla*) — a long crack in the rock where the ground can suddenly move

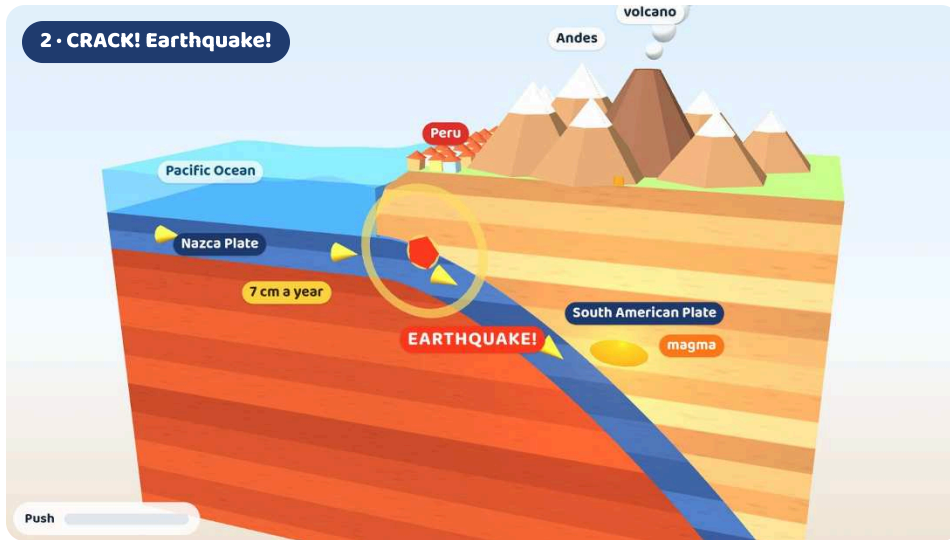
“It's a small earthquake,” says the teacher. “Stay calm.”  
After ten seconds, the ground stops. Everybody is OK.

## A giant puzzle

The Earth is like a big puzzle. The pieces are very, very big. They are called **plates**.

Peru is on a plate. The sea next to Peru is on a plate too. It is called the **Nazca Plate**.





*The Nazca Plate goes under Peru. It pushes... then CRACK! An earthquake.*  
3D model — watch it move in the web edition.

The Nazca Plate goes under Peru. It moves a little every year.

The plates push and push. Then — crack! — they move. The ground shakes. That is an earthquake.

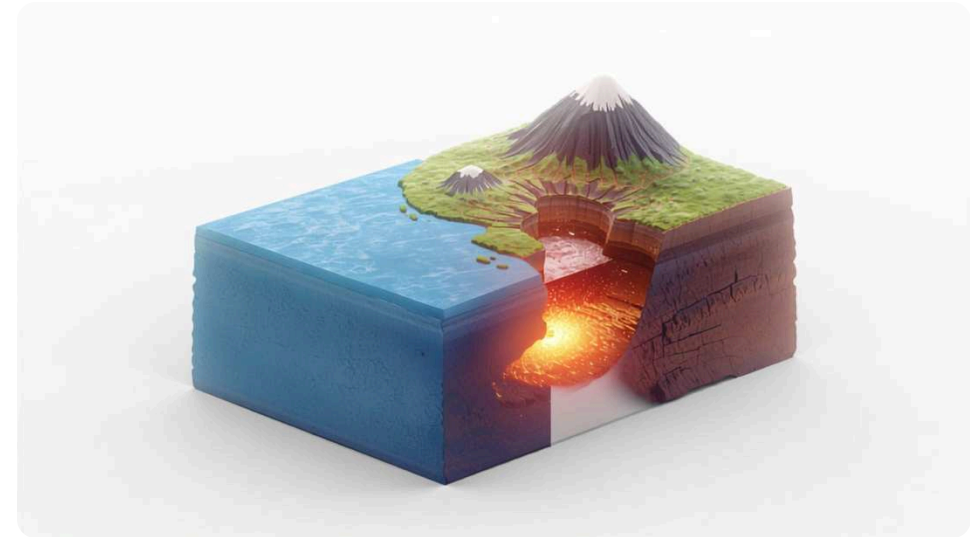
## Timeline of the World

- 1556** —China: the worst earthquake ever.
- 1755** —Portugal: an earthquake and a tsunami in Lisbon.
- 1906** —USA: an earthquake in San Francisco.
- 1912** —Alfred Wegener says: the land moves!
- 1935** —Charles Richter makes his scale.
- 1960** —Chile: the biggest earthquake ever measured.
- 1964** —USA: a big earthquake in Alaska.
- 2004** —Indonesia: a very big tsunami.
- 2010** —Chile: a big earthquake.
- 2011** —Japan: an earthquake and a tsunami.
- 2015** —Nepal: an earthquake near Mount Everest.
- 2023** —Türkiye and Syria: a big earthquake.
- 2025** —Russia: a big earthquake. Waves cross the Pacific.



# Timeline of Peru's Earthquakes, Tsunamis and Volcanoes

- 1582** —An earthquake breaks Arequipa.
- 1600** —The volcano Huaynaputina explodes.
- 1604** —An earthquake and a tsunami in the south.
- 1687** —An earthquake in Lima.
- 1746** —A big earthquake and tsunami in Lima and Callao.
- 1868** —A tsunami puts a ship on the sand in Arica.
- 1922** —The IGP starts in Huancayo.
- 1940** —An earthquake in Lima.
- 1970** —The Áncash earthquake. An avalanche covers Yungay.
- 1972** —Peru makes INDECI.
- 1974** —An earthquake in Lima and Callao.
- 2001** —An earthquake in Arequipa. A tsunami in Camaná.
- 2007** —The Pisco earthquake.
- 2016** —Sabancaya volcano starts to erupt.
- 2019** —A big earthquake in the Amazon.
- 2025** —The IGP writes about 840 earthquake reports.
- 2026** —Millions of children do the national drill.



*Under the sea, one plate goes under the other.*

## The Ring of Fire

Look at the map. There is a big ring around the Pacific Ocean. It is the **Ring of Fire**. Peru is on the Ring of Fire. There are a lot of earthquakes and volcanoes there.



*The Ring of Fire. Find Peru!*

3D model — interactive in the web edition.

## The Puzzle Man

Alfred Wegener was a scientist from Germany. In 1912 he looked at a map. He saw that South America and Africa fit together, like two puzzle pieces! He said: “The land moves.” Many people said, “No!” But he was right.



Photo: Unknown photographer · Public domain



Students in Lima practise an earthquake drill.

Photo: Joaquin Principe · CC BY-SA 4.0



In Peru, this sign shows a safe place to go in an earthquake.

Photo: AgainErick · CC BY-SA 3.0 (also GFDL)



This sign in Lima shows which way to go if a tsunami comes.

Photo: Gaboct · CC BY-SA 4.0

## Two every day

In Peru there are a lot of earthquakes. In 2025 there were more than 800! That is two every day. Most of them are very small.



**LOOK!**

Nobody can say when an earthquake will come. So we must be ready!

## The words for what the Earth does



**earthquake**

The ground shakes.



**tsunami**

A very big wave from the sea.



**landslide**

Rocks and mud fall down a hill.



**flood**

Water covers the streets.

## Animals and Earthquakes

Some people say animals know before an earthquake. Dogs bark. Chickens run. But scientists say: "We don't know!"



## Now you try!

Chapter 1 · Our Ground Moves

### A Draw the four words. Hand in · W1

Draw a picture for each word. Then copy the word.

earthquake

tsunami

landslide

flood

### B Match the words and the meanings.

1. earthquake

a. a big wave

2. tsunami

b. water on the streets

3. landslide

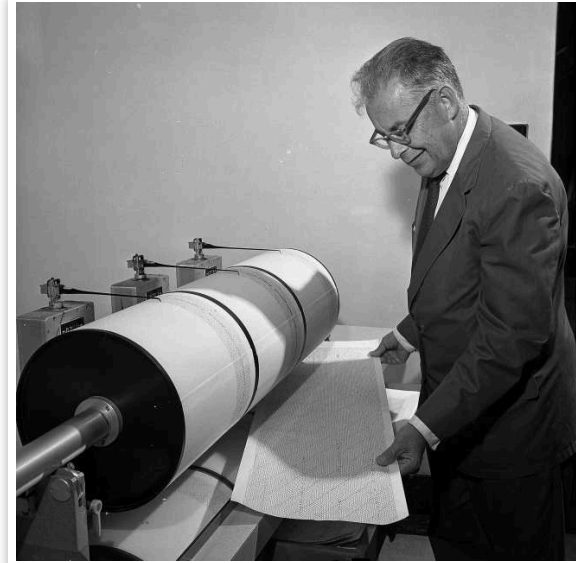
c. the ground shakes

4. flood

d. rocks fall down a hill

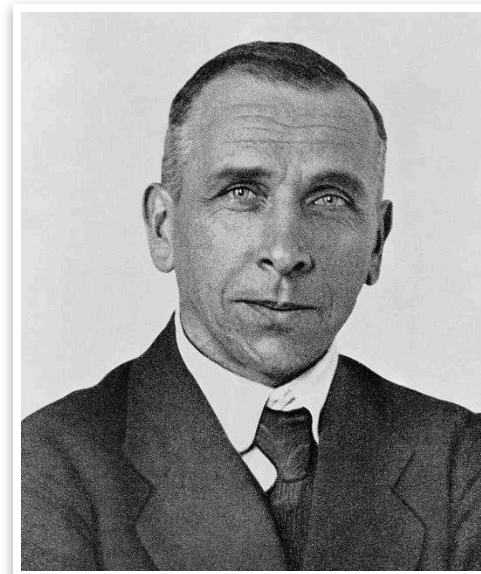
### C Yes or no?

1. Peru is on the Ring of Fire. **yes / no**
2. The Earth is one big piece. **yes / no**
3. In Peru there are earthquakes every day. **yes / no**
4. We know when the next earthquake comes. **yes / no**



Charles Richter made a scale to tell how strong an earthquake is.

Photo: Gil Cooper, Los Angeles Times · CC BY 4.0



Alfred Wegener said that the continents move very slowly.

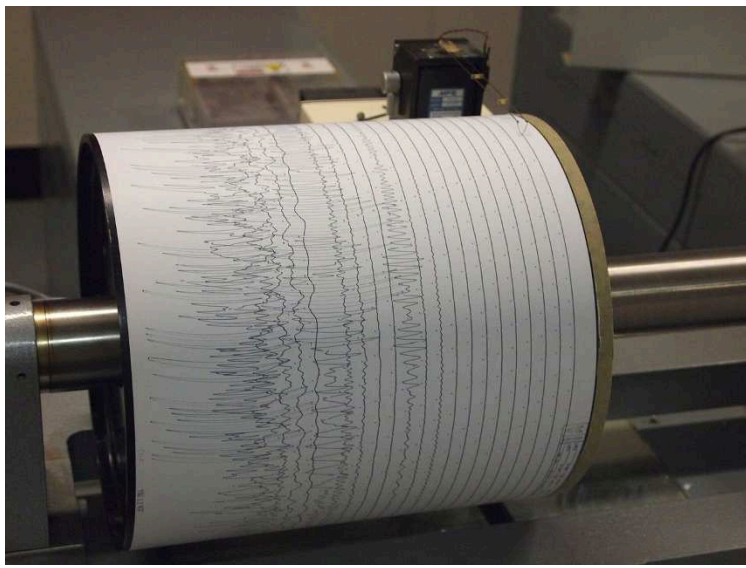
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In 2006, Ubinas volcano blew out a cloud of ash. This photo was taken from space.

Photo: NASA astronaut aboard the International Space Station · Public domain (NASA)



A seismograph draws wiggly lines when the ground shakes.

Photo: Z22 · CC BY-SA 3.0

**D** Look at the picture of the plates. Write the words.

Words: sea · Peru · plate · Andes

1 \_\_\_\_\_ 2 \_\_\_\_\_ 3 \_\_\_\_\_ 4 \_\_\_\_\_

**E** Have you felt one? Ask five friends.

Ask: “Do you feel earthquakes?” Tick ✓ yes or ✗ no.

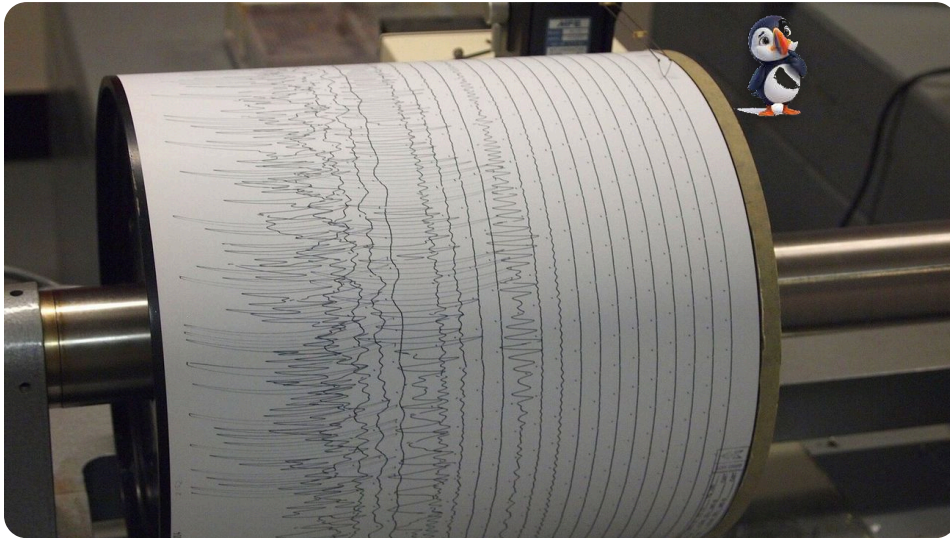
| Name | Yes / No |
|------|----------|
|      |          |
|      |          |
|      |          |
|      |          |

**F** Count!

Two earthquakes every day. How many in one week?  $2 + 2 + 2 + 2 + 2 + 2 + 2$   
 = \_\_\_\_\_



## Reading the Numbers



*A seismograph draws the shaking of the ground as a zig-zag line.*

Photo: Z22 · CC BY-SA 3.0



**Look at the table. Read the numbers with me!**

Scientists have a special machine. It is a **seismograph**.  
When the ground shakes, the machine draws a line. A big  
line is a big earthquake.



**El Misti volcano stands over the city of Arequipa.**

Photo: Josep M. Gracia · CC BY-SA 4.0



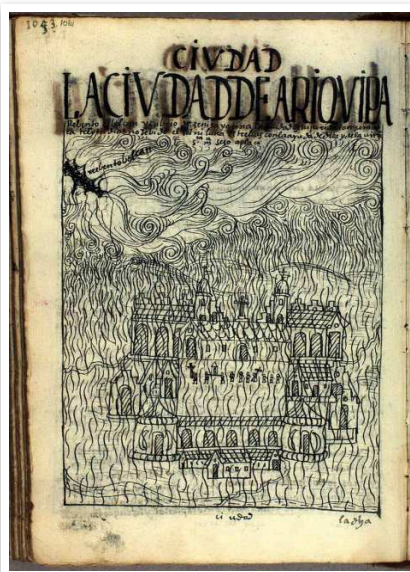
**Sabancaya volcano sent up a big cloud of ash in 2017.**

Photo: Ministerio de Defensa del Perú · CC BY 2.0



Helpers unloaded boxes of supplies for families after the 2007 earthquake.

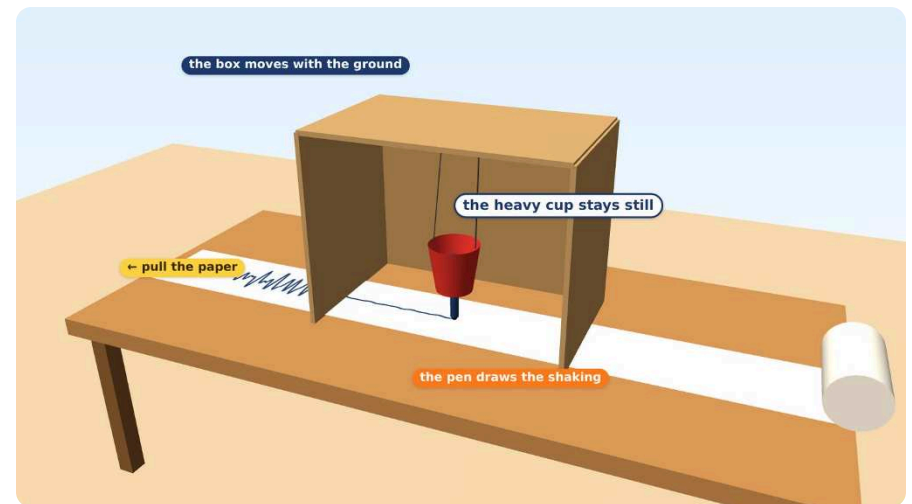
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This old drawing shows ash falling on Arequipa when Huaynaputina erupted in 1600.

Photo: Felipe Guaman Poma de Ayala · Public domain

## Build your own seismograph



### You need

- a shoebox
- a paper cup
- string
- a marker
- coins
- a long strip of paper

### What to do

1. Put the box on its side.
2. Make two holes in the top. Hang the cup with the string.
3. Put coins in the cup. Now it is heavy.
4. Make a hole in the cup. Put the marker in. It touches the paper.
5. Pull the paper slowly. Your friend shakes the table. Look at the line!

**The box moves. The cup stays still. The pen draws the shaking!**



## Richter Scale

Charles Richter was a scientist in the USA. In 1935 he made numbers for earthquakes. A 7 is very strong. An 8 is stronger. A 9 is the strongest!



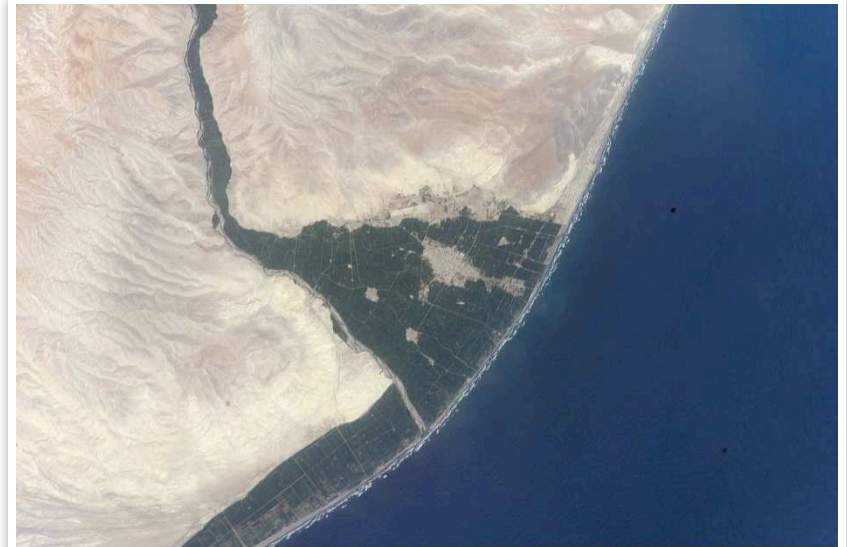
Photo: Gil Cooper, Los Angeles Times · CC BY 4.0

## Magnitude or Intensity?

Magnitude is **one** number for the whole earthquake. **Intensity** is how much you feel it where you are. You feel it more near the **epicentre**. After a big earthquake, small earthquakes can come. They are **aftershocks**.

## Five earthquakes Peru remembers

Peru has earthquakes every day. But five were very, very big. Look at the table.



Astronauts took this photo of Camaná, where a tsunami came on land in 2001.

Photo: NASA Earth Sciences and Image Analysis Laboratory, Johnson Space Center · Public domain (NASA)



After the earthquake of 2007, broken bricks filled the streets of Pisco.

Photo: Senior Airman Shaun Emery, U.S. Air Force · Public domain (U.S. military)



Today the white statue in Yungay looks out at Huascarán.

Photo: Jgaray2000 · CC BY-SA 4.0



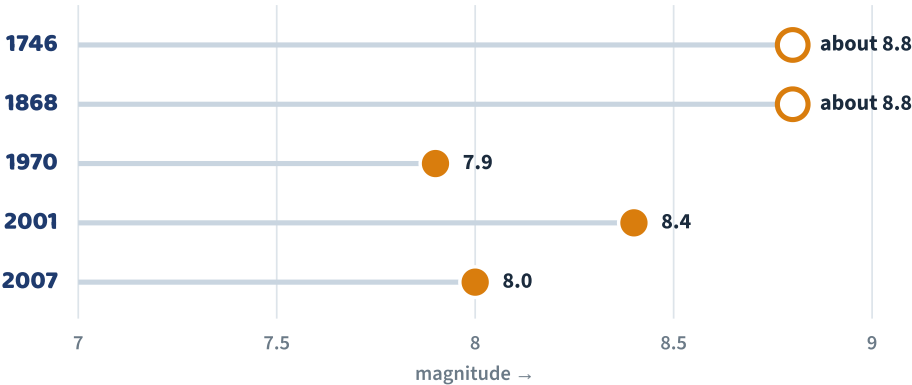
These palm trees grew in the main square of old Yungay. They are still there today.

Photo: Yhhue91 · CC BY-SA 4.0

### Five earthquakes Peru remembers

| Year | Where?                    | Magnitude | Tsunami? |
|------|---------------------------|-----------|----------|
| 1746 | Lima and Callao           | about 8.8 | yes 🌊    |
| 1868 | Arica                     | about 8.8 | yes 🌊    |
| 1970 | Áncash                    | 7.9       | no       |
| 2001 | Arequipa, Moquegua, Tacna | 8.4       | yes 🌊    |
| 2007 | Pisco, Ica, Chincha       | 8.0       | small    |

Which is the biggest number? Which earthquake has a tsunami?



Magnitude of the five earthquakes. The big numbers are the strongest.

### The IGP: Peru's Earthquake Watchers

The IGP is in Peru. The scientists watch the ground every day and every night.





### LOOK!

Magnitude 9.5 — Chile, 1960 — is the biggest earthquake ever measured.

## Asking about data

Ask your friend: **When was the big earthquake in Pisco?**

— “In 2007.”



After the avalanche of 1970, the statue on the cemetery hill in Yungay was still standing.

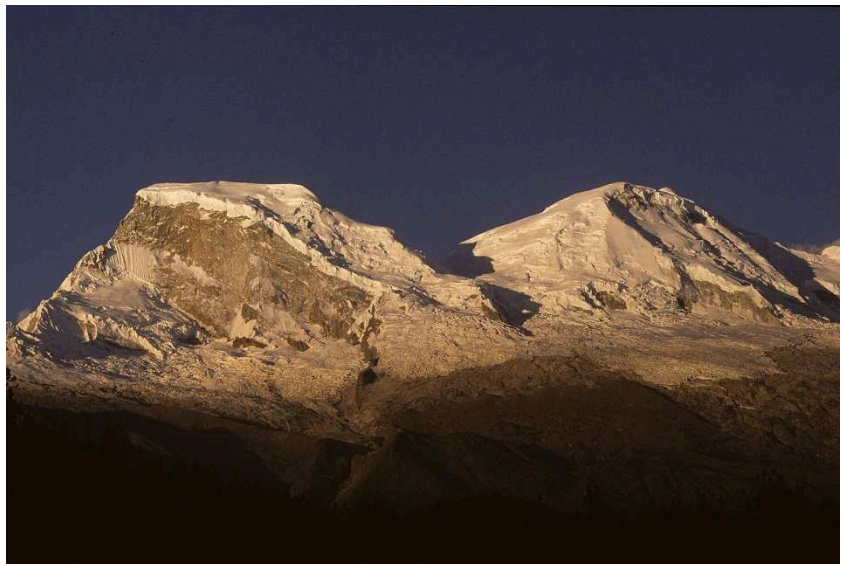
Photo: U.S. Geological Survey · Public domain (USGS)



In 1980, people built this gate at the Yungay memorial. Huascarán is behind it.

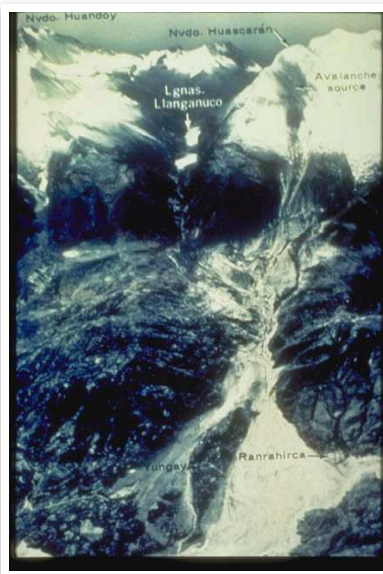
Photo: DB · Public domain (released by the author)





Huascarán is the highest mountain in Peru.

Photo: Maurice Chédel · Public domain (released by the author)



From the sky you can see the path of the 1970 avalanche from Huascarán down to Yungay.

Photo: George Plafker, U.S. Geological Survey · Public domain (USGS / NOAA)

## Now you try!

Chapter 2 · Reading the Numbers

### A Copy five true sentences. Hand in · W2

Circle the true sentences. Copy them.

1. The Pisco earthquake was in 2007.
2. The Áncash earthquake was in 1970.
3. There was a tsunami in Arica.
4. The 2001 earthquake was in Lima.
5. The 1746 earthquake was in Lima and Callao.
6. There was a big tsunami in 1970.

### B Read the table and answer.

1. When was the Pisco earthquake? \_\_\_\_\_
2. Where was the 1868 earthquake? \_\_\_\_\_
3. How many earthquakes are in the table? \_\_\_\_\_

### C Big, bigger...

8.8 is **bigger** than 7.9. ✓ 7.9 is bigger than 8.4. ✗

Now you: 8.4 is \_\_\_\_\_ than 8.0.

### D Read the chart.

Look at the magnitude chart. Colour the strongest earthquakes red.

**E Ten times!**

A magnitude 5 makes waves 10 times bigger than a 4. Draw a small wave and a big wave.

**F Speaking: data detectives**

Point to the table. Say: “1970. Áncash. 7.9.” Your friend says the right row.



Parts of the Wateree are still on the beach near Arica today.

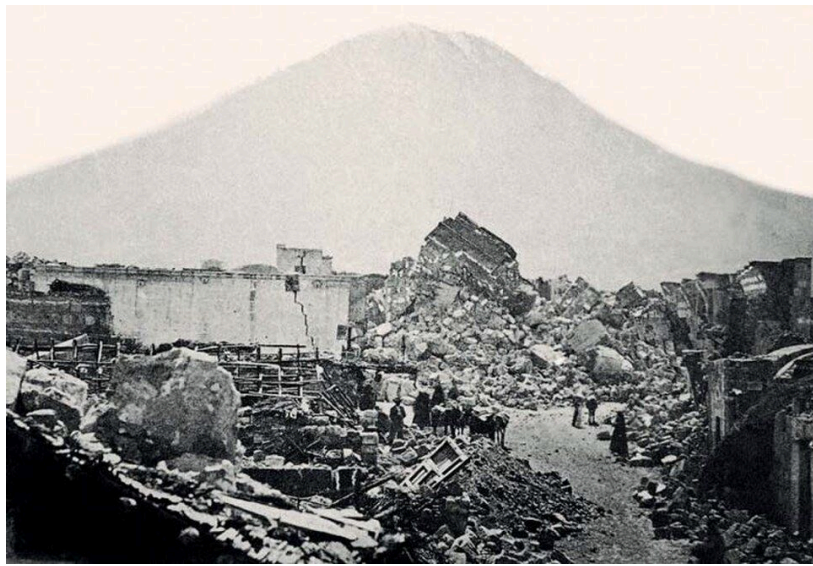
Photo: Pinot · Public domain (released by the author)



The Morro de Arica is a tall rock hill next to the sea.

Photo: Jowyto · CC BY-SA 3.0





The earthquake of 1868 knocked down many buildings in Arequipa.

Photo: Unknown photographer, c. 1868 · Public domain



The tsunami of 1868 carried the ship Wateree onto the land at Arica.

Photo: Unknown photographer, 1868; Henry Clay Cochrane Collection, USMC Archives · CC BY 2.0

## CHAPTER 3

W3 · Construcción

# What Happened in 1970



The big mountain Huascarán and the town of Yungay.



What happened in Yungay? Let's read.

It is Sunday, 31 May 1970. It is a sunny day in Áncash. There are big mountains. The highest mountain is **Huascarán**. It has a lot of snow.

In the town of Yungay, people are happy. Some families watch football on TV. Children go to the circus.

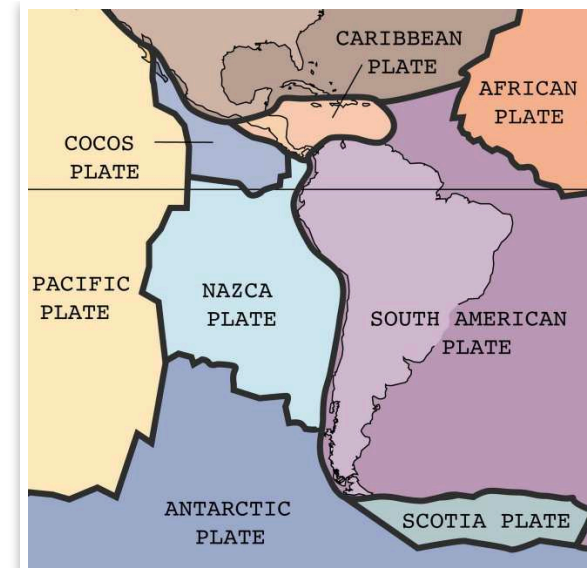
At 3:23 in the afternoon, the ground shakes. It shakes for 45 seconds. Many houses fall.



*Ice and rocks fall from the mountain.*

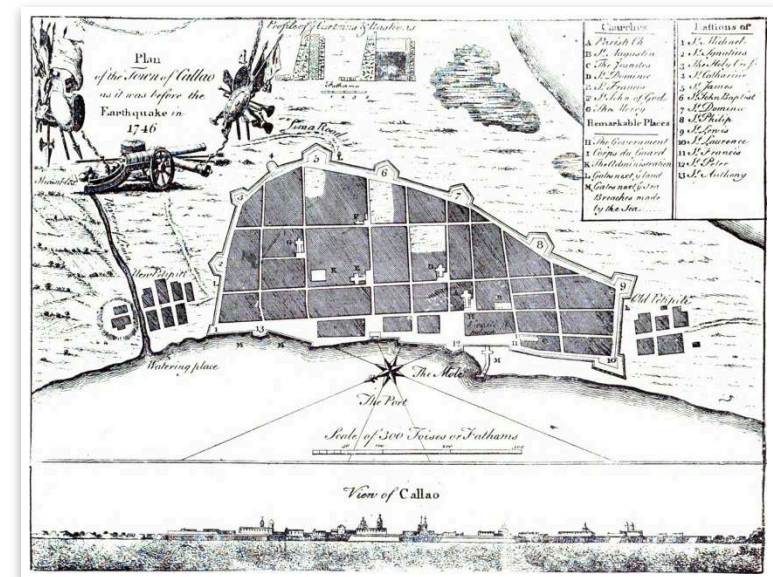
Then, on the mountain, a very big piece of ice and rock falls. It goes down very, very fast. It covers the town of Yungay with mud and rocks.

Some people run up a hill. It is the cemetery. They are safe. The children at the circus are safe too. But many, many people die. It is very, very sad.



Peru is where the Nazca Plate and the South American Plate meet.

Photo: U.S. Geological Survey · Public domain (USGS)



This old map shows the port of Callao before the earthquake and tsunami of 1746.

Photo: Unknown 18th-century engraver; reproduced by The West Coast Leader, Ltd. · Public domain



# Peru in Photos

Real photos of earthquakes and volcanoes in Peru.



Many old houses in Lima have wooden balconies.

Photo: Felipe Restrepo Acosta · CC BY-SA 4.0

## The Four Palm Trees

After the avalanche, you can see the tops of four palm trees. They are still there today!



Photo: Yhhue91 · CC BY-SA 4.0



*Yungay today. It is a quiet, green park.*

## After the earthquake

People from all of Peru help. People from other countries help too. They bring food, water and tents.



### Astrid's tip

**It is 1970.** We can tell a story in the present: *The ground shakes. The ice falls. People run.*

## Now you try!

Chapter 12 · The Safety Fair

### A Your presentation and the family reflection.

Hand in · W11

Say: "Hello! This is my flyer. Before an earthquake... During... After..."

Then write about your home:

We have \_\_\_\_\_. We don't have \_\_\_\_\_. I can \_\_\_\_\_.

### B Questions visitors might ask

Practise: "What's in the backpack?" — "Water, a torch..."

### C How did I do? (Rubric)

Colour the stars: ★★★★★ I know the words · ★★★★★ I can say what to do · ★★★★★ I made my flyer



### Presenting my flyer

1. Hello! My name is .... This is my flyer.
2. Before an earthquake, ....
3. During an earthquake, ....
4. After an earthquake, ....
5. Thank you! Any questions?

### And in my house?

Look at your home. Do you have a backpack? Do you have water? Write one thing you can do.

| At home                | We have it ✓             | We don't have it ✗       |
|------------------------|--------------------------|--------------------------|
| emergency backpack     | <input type="checkbox"/> | <input type="checkbox"/> |
| water                  | <input type="checkbox"/> | <input type="checkbox"/> |
| torch and batteries    | <input type="checkbox"/> | <input type="checkbox"/> |
| whistle                | <input type="checkbox"/> | <input type="checkbox"/> |
| first-aid kit          | <input type="checkbox"/> | <input type="checkbox"/> |
| a family meeting point | <input type="checkbox"/> | <input type="checkbox"/> |

Remember the big question? Now you know the answer.  
You are ready!

## Now you try!

### A Your report about 1970 — four sentences.

Hand in · W3

Complete and copy.

1. It is Sunday, 31 May \_\_\_\_\_.
2. The ground \_\_\_\_\_.
3. Ice and rocks \_\_\_\_\_ from Huascarán.
4. People \_\_\_\_\_ up the hill.

### B Put the events in order (1–4).

- ☐ Ice and rocks fall.
- ☐ It is a sunny Sunday.
- ☐ The ground shakes.
- ☐ People go up the hill.

### C Find the words.

Circle: **mountain** · ice · pizza · town · snow · car — Which words are in the story?

### D Then and now

Look at the two pictures of Yungay. Say one thing: “Now there is a park.”



**E A message of solidarity**

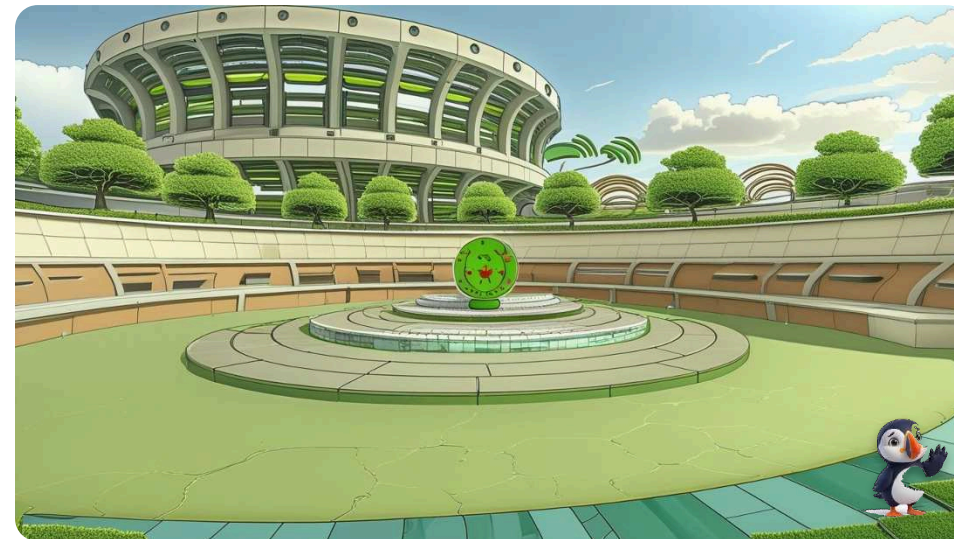
Draw a heart and write: **We remember Yungay.**

---

**CHAPTER 12**

W11 · Producto y audiencia

## The Safety Fair



*The safety fair at school.*



**We did it! Our flyers are ready!**

Today is the safety fair. Families come to school. You show your flyer. You say what to do.

## Now you try!

Chapter 11 · Making It Better

### A Your finished flyer. Hand in · W10

Copy your flyer on a big paper. Use colours. Draw three pictures.

### B Two stars and a wish

★ I like... ★ I like... ✂ Change...

### C Fix the flyer.

Find 3 mistakes in Nico's flyer (in the box above).

## CHAPTER 4

W4 · Borrador

# When the Sea Comes



*A tsunami is a very big wave.*



**Oops! Is a tsunami one wave? No — many waves!**

It is night. It is the year 1746. The ground in Lima shakes. Then the sea comes. A very big wave covers the port of Callao.

## What is a tsunami?

An earthquake under the sea moves the water. The water makes big waves. They are called a **tsunami**. The waves are very fast.

### A Japanese Word

“Tsunami” is a Japanese word. It means “harbour wave”.

## The ship in the desert

In 1868, in Arica, a big tsunami comes. It carries a ship onto the land! The ship is on the sand, far from the sea.



*A ship on the sand!*

### ✓ Checklist

- ☐ My flyer has a BEFORE part.
- ☐ My flyer has a DURING part.
- ☐ My flyer has an AFTER part.
- ☐ I use must, must not or should.
- ☐ I say what the family must have ready.
- ☐ I say where to go.

## Check the verbs

Instructions start with a verb: **Pack** a torch. **Go** up. **Don't** run.



### Nico's flyer — find the mistakes!

- BEFORE: Pack a backpack.
- DURING: Run outside!
- AFTER: Go to the beach.
- Cover your feet.

## Read it out loud

Read your flyer to the class. Speak loud. Speak slowly. Look at the people.



# Making It Better



We check our flyers together.



Oops! My flyer has mistakes. Can you help me?

Swap flyers with a friend. Read it. Use the checklist. Say two good things and one thing to change.

## The earthquake is the alarm

Are you near the sea? Does the ground shake a lot? Go up!  
Go to a high place. Don't wait.

### Five steps

- 1  The ground shakes a lot. It is your alarm.
- 2  Walk. Go away from the beach.
- 3  Go up! Go to a high place.
- 4  Stay there. More waves come.
- 5  Go back when adults say “OK”.

### A Wave from Far Away

In 2025 there was a big earthquake in Russia. Peru closed the beaches. The waves were small.



Photo: Gaboct · CC BY-SA 4.0

## Now you try!

Chapter 4 · When the Sea Comes

### A The five steps — in order. Hand in · W4

Number the pictures and sentences 1–5. Copy them in your notebook.

- ☐ Stay there.
- ☐ The ground shakes a lot.
- ☐ Go up!
- ☐ Go back when adults say OK.
- ☐ Walk away from the beach.

### B Tsunami: true or false?

1. A tsunami is a big wave. **yes / no**
2. We go to the beach to look. **yes / no**
3. We go to a high place. **yes / no**

### C Read the signs.

Look at the green sign with the arrow. What does it say? Colour an arrow green.

### D The ship in the desert — questions

What is on the sand? A \_\_\_\_\_.

### E Role-play: on the beach

Play in pairs. One says: “The ground shakes!” The other says: “Go up!” and walks up the stairs.

## Now you try!

Chapter 10 · Writing My Flyer

### A Your first draft, with its three drawings. Hand in · W9

Complete the flyer frame. Then draw one picture in each box.

#### BEFORE — Get ready!

Pack a backpack. Put in water, a \_\_\_\_\_, a \_\_\_\_\_ and a \_\_\_\_\_. Keep it by the \_\_\_\_\_.

draw

#### DURING — Drop, cover, hold on!

Don't \_\_\_\_\_. Go down on your \_\_\_\_\_. Cover your \_\_\_\_\_. Hold on to a \_\_\_\_\_.

draw

#### AFTER — Go out!

Walk to the \_\_\_\_\_ point. Don't run, don't shout, don't \_\_\_\_\_. Near the sea? Go \_\_\_\_\_!

draw

### B must or must not?

You \_\_\_\_\_ pack water. You \_\_\_\_\_ run on the stairs.

| BEFORE               | DURING               | AFTER                |
|----------------------|----------------------|----------------------|
| Heading:<br><br>     | Heading:<br><br>     | Heading:<br><br>     |
| Instruction:<br><br> | Instruction:<br><br> | Instruction:<br><br> |
| Picture:<br><br>     | Picture:<br><br>     | Picture:<br><br>     |

Word count: \_\_\_\_\_ (90–110)

Word bank

- Disasters** earthquake · tsunami · landslide · flood · volcano · ash
- Safety** safe zone · drill · exit · meeting point · shelter · danger · Drop, cover, hold on
- Backpack** emergency kit · water · torch · whistle · blanket · first-aid kit · radio
- Grammar** must · must not · before · during · after

Sentence starters

Pack a... · Keep your... · Drop, cover and... · Don't run! · Go to the... · Call...



**LOOK!**  
Count your words! 90 is a lot — but the frames help you.

CHAPTER 5

Science link · Reading plan

Mountains of Fire



The volcano El Misti and the city of Arequipa.



What's this? A mountain with smoke!

A **volcano** is a mountain. Inside, there is very hot rock. It is called **lava**. Sometimes smoke and ash come out of the top.



*Hot rock goes up inside the volcano.*

Peru has a lot of volcanoes. They are in the south of Peru.

### The biggest eruption in South America

In the year 1600, a volcano in Moquegua explodes. Its name is **Huaynaputina**. There is a lot of ash. The ash covers the city of Arequipa.

## CHAPTER 10

W9 · Producto y audiencia

# Writing My Flyer



*Time to write!*



**First plan. Then write. Then draw!**

Now you make your flyer. It has three parts: **BEFORE**, **DURING**, **AFTER**. Each part has a heading, an instruction and a picture.



## Now you try!

Chapter 9 · How a Safety Flyer Works

### A Your three headings. Hand in · W8

Copy and complete:

**BEFORE** — Pack a \_\_\_\_\_.

**DURING** — Drop, cover, \_\_\_\_\_.

**AFTER** — Go to the \_\_\_\_\_.

### B Label the model flyer.

Find on the flyer: the **title**, a **picture**, a **heading**. Point and say.

### C Find the hidden instruction.

“The backpack is by the door.” → Put the backpack \_\_\_\_\_ the door.

### D Good flyer or bad flyer?

Which is better? A: “DROP! COVER! HOLD ON!” B: “It is important that people think about what they can do.”



*A volcano erupts. The llamas run away!*

### The White City

Arequipa is the “White City”. Many houses are made of white volcano stone!



Photo: Josep M. Gracia · CC BY-SA 4.0

Some volcanoes in Peru are active today. **Sabancaya** has ash every day!



**LOOK!**

A volcano can be higher than 5,000 metres!

## Now you try!

Chapter 5 · Mountains of Fire

### A Label the volcano.

Draw a volcano. Write: **top · smoke · lava**

### B Big, bigger, biggest

Sabancaya 5,960 m · El Misti 5,822 m · Ubinas 5,672 m

The highest volcano is \_\_\_\_\_.

### C True or false?

1. Peru has volcanoes. **yes / no**
2. The volcanoes are in the north. **yes / no**

### D Make a volcano! (with your teacher)

You need: a bottle, baking soda, vinegar, red colour. Put in the soda. Pour in the vinegar. Look!

## Is your family ready?

### BEFORE Get ready!



Pack a backpack: water, a torch and a whistle. Put it by the door.

*Our backpack is by the door.*

### DURING Drop, cover, hold on!



Don't run. Go down. Cover your head. Hold on.

*Stay under the table.*

### AFTER Go out!



Walk to the meeting point. Don't run, don't shout, don't push. Near the sea? Go up!

*Our meeting point is the park.*

Emergency: 116 · 105 · 106 · Voice messages: 119

Model flyer — 70 words.

## The parts of a flyer



### Title

The big words at the top.



### Heading

The name of each part.



### Instruction

It says: do this!



### Picture

It shows the action.



### Caption

Small words under a picture.

## The hidden instruction

“A whistle is very loud.” This means: **Pack a whistle!**

## CHAPTER 9

W8 · Producto y audiencia

# How a Safety Flyer Works



*Signs are short and clear.*

Photo: Gaboct · CC BY-SA 4.0



What's this? It's a flyer! Big words, big pictures.

A **flyer** is one page. It has a title, big pictures and short sentences. It tells people what to do.

## CHAPTER 6

W5 · Final y entrega

# When People Get Hurt



*Pisco, 2007. Many houses fall.*

Photo: Senior Airman Shaun Emery, U.S. Air Force · Public domain (U.S. military)



My knee hurts! Who can help me?

It is 15 August 2007. It is 6:40 in the evening. In Pisco, the ground shakes for a long time. Many houses fall. Many people are hurt.

Firefighters help. Doctors help. People from all of Peru bring food and water.





*Planes bring help.*

Photo: Tech. Sgt. Jeremy Lock, U.S. Air Force · Public domain (U.S. military)

### **E** Before, during or after?

Draw a line to **before, during** or **after**: Drop, cover, hold on · Pack the backpack · Go to the meeting point.

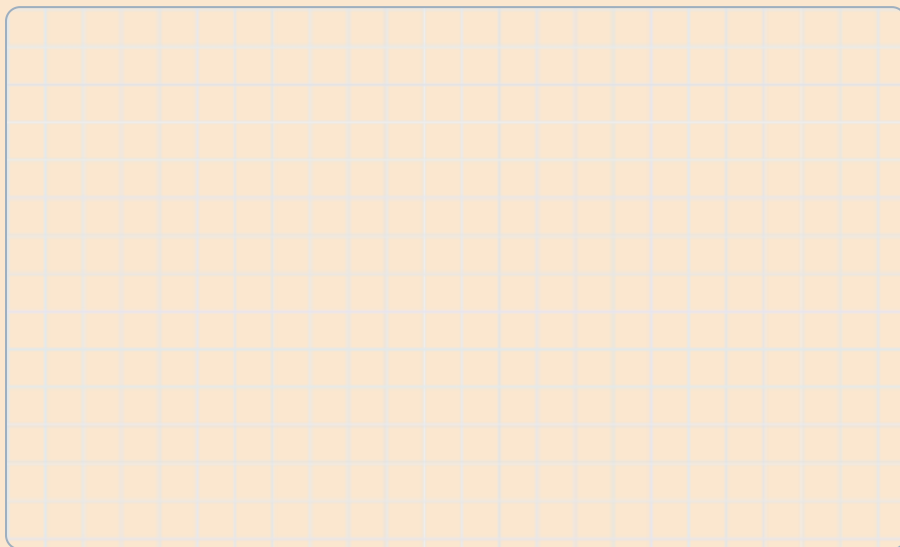


## Now you try!

Chapter 8 · The Safe Zone and the Drill

### A Draw your classroom. Hand in · W7

Draw your classroom from above. Draw the door. Colour the **safe zone** green.



### B Put the drill in order.

☐ Walk out. ☐ Drop, cover, hold on. ☐ The teacher counts.

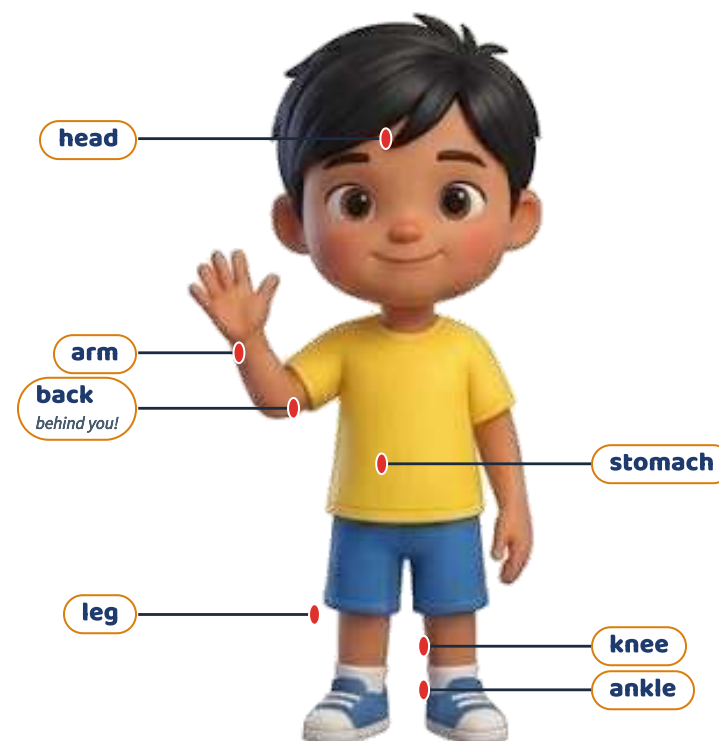
### C Game: Simon says

Your teacher says “Simon says drop!” — drop! If the teacher doesn't say “Simon says”, don't move!

### D Say the chant.

*Drop, drop, drop! Cover, cover, cover! Hold on, hold on, hold on — until it stops!*

## My body



*Point to your head, arm, back, leg, knee and ankle!*

**Where does it hurt?** — My arm hurts. · My leg hurts. · My head hurts.

**I have a headache.** (head + ache)

**Who do you tell? What do you say?**

Tell an adult: a teacher, your mum or your dad. Say:  
“Help, please! My friend is hurt.”



**116** Firefighters  
*Bomberos*



**105** Police  
*Policía*



**106** Ambulance  
*SAMU*



**115** Civil defence  
*Defensa Civil*



**119** Leave a voice message  
*Mensajes de voz*

### Line 119: Leave a Message

After a big earthquake, phones can be busy. Call 119 and leave a message for your family!



**Freya:** Are you OK?



**Nico:** No. My arm hurts.



**Freya:** Sit here. I can call the teacher.

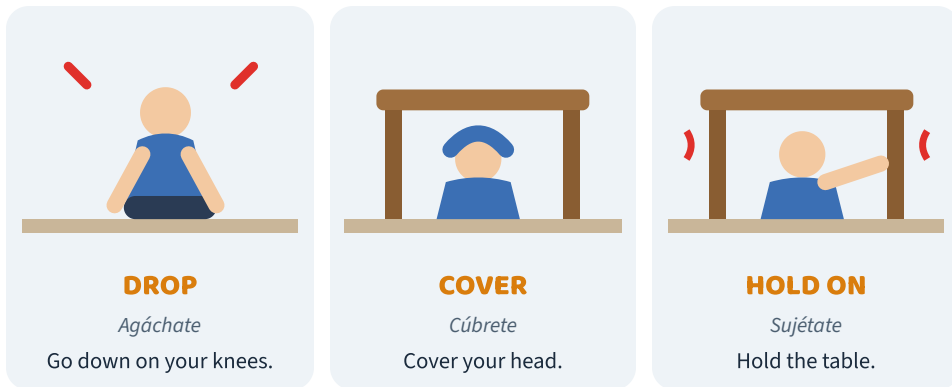


**Nico:** Thank you!

### No corro, no grito, no empujo

In Peru, children say: **No corro, no grito, no empujo.** I don't run. I don't shout. I don't push.

## Drop, cover, hold on



**Drop:** go down on your knees. **Cover:** cover your head.  
**Hold on:** hold the table.

## The three moments of a Peruvian drill

### The drill

- 1 The alarm rings. Drop, cover, hold on.
- 2 Walk out. Don't run. Don't shout. Don't push.
- 3 Go to the meeting point. The teacher counts.

Peru has a big drill for all the country. It is the **Simulacro Nacional**. Millions of children do it!

## Now you try!

### A A four-line dialogue: somebody is hurt and you help.

Hand in · W5

Complete. Then act it out.

A: Are you OK?

B: No. My \_\_\_\_\_ hurts.

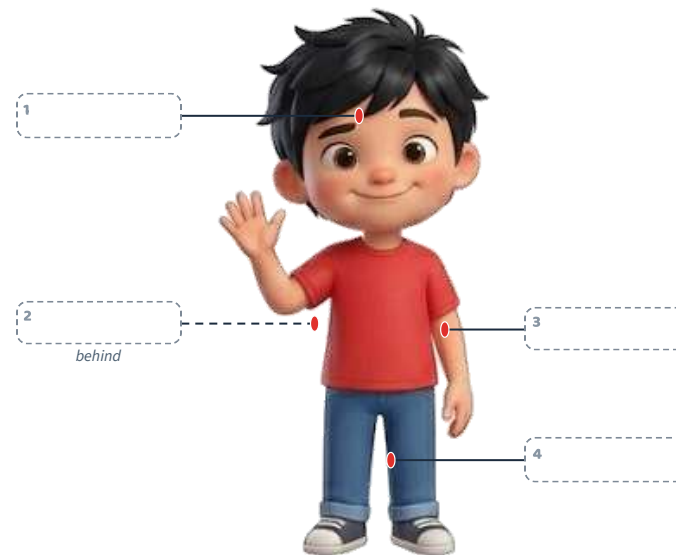
A: Sit here. I can call the \_\_\_\_\_.

B: \_\_\_\_\_!

### B Label the body.

Put the words in the boxes.

Words: **head** • **back** • **arm** • **leg**



### C What's the problem?

Match.

1. headache

a. 🤒 head

2. stomachache

b. 🤮 stomach

3. backache

c. 🧑 back

### D Emergency numbers quiz

Firefighters: 1 \_\_\_ \_\_\_ Ambulance: 1 \_\_\_ \_\_\_

### E Who, where, what?

Look at the picture of Tomás. Say: "Tomás is hurt. He is in the playground."



*A green sign: Zona Segura.*

Photo: AgainErick · CC BY-SA 3.0 (also GFDL)



# The Safe Zone and the Drill



A drill at a school in Lima.

Photo: Joaquin Principe · CC BY-SA 4.0



**Drop! Cover! Hold on!**

At school there are green signs. They say **ZONA SEGURA**. It means **safe zone**. In an earthquake, we go there.

# The Backpack



## What is in the backpack?



water (4 × ½ litre)  
*agua embotellada*



torch and batteries  
*linterna y pilas*



whistle  
*silbato*



tinned food  
*comida enlatada*



masks  
*mascarillas*



first-aid kit  
*botiquín*



small radio  
*radio portátil*



blanket  
*manta polar*



biscuits and chocolate  
*galletas y chocolate*



hand gel  
*gel antibacterial*



**My backpack is ready. Is yours?**

After a big earthquake, there is no light. There is no water in the tap. The shops are closed. We need a special bag. It is the **emergency backpack**.

## The real INDECI rules

**24 h**

For 1 day (24 hours).

**8 kg**

Not heavy: 8 kilos.

**1:2**

1 bag for 2 people.

**1 L**

Water: 1 litre for each person.

## What is in the backpack?

|                                                                                                                                  |                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
|  water (4 × ½ litre)<br><i>agua embotellada</i> |  first-aid kit<br><i>botiquín</i>                      |
|  torch and batteries<br><i>linterna y pilas</i> |  small radio<br><i>radio portátil</i>                  |
|  whistle<br><i>silbato</i>                      |  blanket<br><i>manta polar</i>                         |
|  tinned food<br><i>comida enlatada</i>          |  biscuits and chocolate<br><i>galletas y chocolate</i> |
|  masks<br><i>mascarillas</i>                    |  hand gel<br><i>gel antibacterial</i>                  |

We **must** have water. We **must** have a torch.

We **must not** put toys in the bag. It is heavy!

### Why a Whistle?

A whistle is small. It is loud. People can hear you!

### The Reserve Box

At home there is a big box too. It has food and water for more days.

## Now you try!

Chapter 7 · The Backpack

### A Your list of eight things. Hand in · W6

Draw 8 things for your backpack. Write the words.

|   |   |   |   |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |

### B Yes or no?

Put ✓ or X: water ☐ · torch ☐ · TV ☐ · whistle ☐ · football ☐ · blanket ☐

### C Count!

4 bottles of water. 2 cans. 1 torch. How many things? \_\_\_\_

### D Nico's backpack

Nico's bag has: a TV, a teddy, water, crisps. What is wrong? Tell your friend.

### E Take it home

Show this page to your family. Tick what you have at home.